www.tntextbooks.in

- It bums with luminous sooty flame in contrast to alkanes and alkenes which usually burn with bluish flame.
- Their vapours are highly toxic which an inhalation produce loss of consciousness.

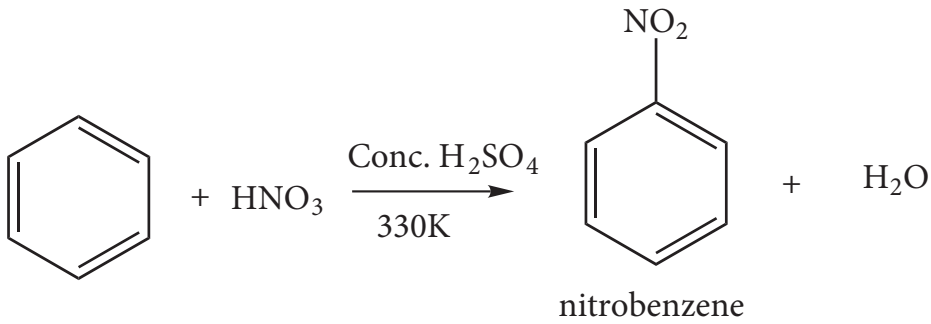
## 13.5. Chemical Properties:

1. Benzene contains delocalized π-electrons which make the ring to act as an electro rich centre. So electrophilic substitution reaction occurs in benzene.
2. Benzene ring is stabilized by delocalized π electrons. Though it is highly stable, it undergoes addition and oxidation reaction under specific conditions.

### 1. Electrophilic Substitution Reaction

### (a) Nitration:

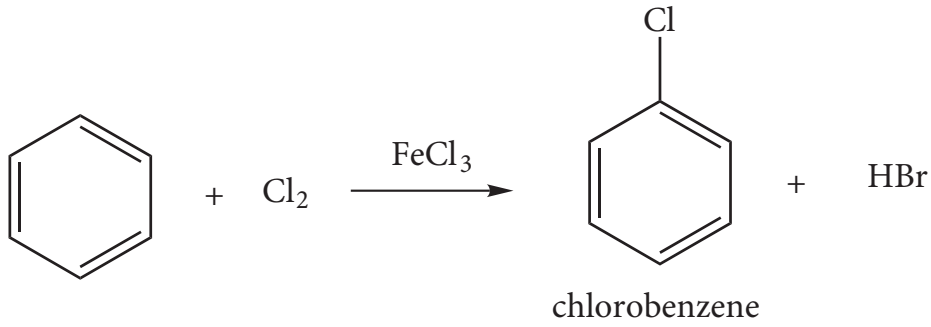
When benzene is heated at 330K with a nitrating mixture (Con. $HNO_3$ + Con. $H_2SO_4$), nitro benzene is formed by replacing are hydrogen atom by nitronium ion $NO^{2+}$(electrophile)




Concentrated $H_2SO_4$ is added to produce nitronium ion $NO^{2+}$

### (b) Halogenation:

Benzene reacts with halogens ($X_2$=$Cl_2$, $Br_2$,) in the presence of Lewis acid such as $FeCl_3$, $FeBr_3$ or $AlCl_3$ and give corresponding halo benzene. In the absence of catalyst, Fluorine reactsvigoursly with benzene even in the absence of catalyst. However iodine is very inactive even in the presence of catalyst




### (c) Sulphonation:

Benzene reacts with fuming sulphuric acid (Con $H_2SO_4$ + $SO_3$) and gives benzene sulphonic acid. The electrophile $SO_3$ is a molecule. Although it does not have positive charge, it is a strong electrophile. This is because the octet of electron around the sulphur atom is not reached. The reaction is reversible and desulphonation occurs readily in aqueous medium.